Invariant Geodesic Trajectory for Neural Machine Translation: Flow Matching on Sequence Manifolds

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Abstract

We introduce the **Topo-Geometric Flow Invariant** (TGFI) framework, realised as **Geodesic Flow Matching on Sequence Manifolds** (GFM-SM), a principled reformulation of neural machine translation as continuous boundary-value mechanics on a Poincaré ball manifold. Rather than mapping discrete token sequences, TGFI constructs a piecewise geodesic trajectory on the Poincaré ball from source token embeddings, compresses it into a length-invariant trajectory bottleneck Ψ via a neural ODE integrated along the trajectory, and conditions an autoregressive tangent-space Transformer through a learned gate. A Riemannian conditional flow matching (CFM) objective trains a velocity field that transports source hyperbolic trajectory waypoints toward corresponding target waypoints, providing an auxiliary cross-lingual geometric regulariser. A projective quantiser maps flowed hyperbolic states back to vocabulary logits via a decode-decoupled training signal. All three components are jointly optimised end-to-end under a unified objective while preserving stable autoregressive inference. Experiments on five benchmark datasets, namely WMT16 EN-DE (NC-v11 and full), IWSLT15 EN-VI, WMT14 EN-DE, WMT16 EN-RO, and IWSLT14 DE-EN, demonstrate consistent improvements over vanilla Transformers, AMR-augmented Transformers, and flow-based non-autoregressive models, all without relying on external semantic parsers [26] or knowledge distillation [8]. We also provide a rigorous analysis of the train/inference mismatch failure mode that arises when auxiliary geometric logits are naively fused at decode time, and show how decode decoupling resolves it.

1 Introduction

Neural machine translation has converged on the Transformer as its dominant architecture [27], with improvements coming largely from scale, pre-training [4, 17], and structural augmentation with external semantic graphs [26, 13]. These approaches share a fundamental assumption: translation is a discrete token-mapping problem, and the geometry of the underlying representation space is an incidental detail rather than a design choice.

We challenge this assumption. Language is hierarchical; sentences have constituent structure that branches exponentially with depth. Hyperbolic geometry, whose volume grows exponentially with radius, mirroring this branching factor [22], is the natural ambient space for source sentence representations. If source token embeddings are lifted into the Poincaré ball and a piecewise geodesic path is traced through them in token order, the resulting trajectory encodes the hierarchical structure of the sentence in a geometrically principled way. The *curvature* of this trajectory reflects compositional organisation that flat Euclidean attention cannot capture directly.

The challenge is using this geometric representation to condition a competitive autoregressive decoder. We achieve this through three jointly trained components: **(i)** a geodesic spline construction that builds the source trajectory on the Poincaré ball and integrates a neural ODE along it to produce the length-invariant trajectory bottleneck Ψ, a fixed-dimensional summary of the sentence's global geometric structure; **(ii)** a Riemannian conditional flow matching objective that trains a velocity field to transport source trajectory waypoints toward target waypoints, regularising the shared hyperbolic encoder; **(iii)** a projective quantiser that maps flowed hyperbolic states to auxiliary vocabulary logits, providing an additional gradient path into the geometric modules during training.

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A central engineering insight is that auxiliary geometric logits *must* be excluded from inference (*decode decoupling*). Including them introduces a systematic train/inference mismatch: the quantiser is trained against teacher-forced target geometry that does not exist at decode time, so its logits become confidently wrong as training progresses, eventually dominating and collapsing translation quality to near-zero BLEU despite a falling training loss. We document this failure mode formally and show that decode decoupling resolves it while retaining the full auxiliary training signal.

Contributions. We introduce TGFI/GFM-SM, the first NMT architecture to jointly train hyperbolic geodesic source trajectories, a length-invariant neural-ODE bottleneck Ψ, Riemannian CFM cross-lingual transport, and projective quantisation in a single end-to-end objective. We provide a rigorous mathematical derivation of every component and five formal algorithms covering training, inference, and the three geometric sub-procedures. We demonstrate consistent improvements over strong autoregressive and non-autoregressive baselines across five benchmarks and four language pairs, without external parsers or knowledge distillation. We give a principled analysis of the decode-decoupling requirement and its connection to the broader class of teacher-forcing / inference mismatch problems.

2 Related Work

Structural augmentation for NMT. Injecting syntactic or semantic structure into NMT has a long history [1, 19, 26]. Li and Flanigan [13] combine a Transformer encoder-decoder with a Heterogeneous Graph Transformer [28] encoding source AMR graphs, achieving consistent gains on WMT16 EN-DE and IWSLT15 EN-VI, which serve as the direct baseline for our experiments. TGFI differs fundamentally: it derives geometric structure entirely from source token embeddings without any parser, graph construction, or auxiliary encoder, while achieving superior or comparable performance at a fraction of the additional parameter cost.

Hyperbolic representation learning. Nickel and Kiela [22] established that Poincaré embeddings can represent hierarchical data with exponentially lower distortion than Euclidean embeddings. Ganea et al. [6] extended this to neural network layers and attention. Dhingra et al. [5] showed advantages in NLP for text with latent hierarchical structure. Most relevant is Li et al. [14], who apply Lorentz-manifold flow matching to cross-modal few-shot adaptation, motivating our use of hyperbolic geometry for cross-lingual trajectory transport. Our setting differs in that trajectories are constructed from variable-length token sequences rather than static embeddings, and the velocity field conditions on an autoregressive decoding prefix.

Flow matching and Riemannian generative models. Flow matching [15, 16] is a simulation-free framework for learning velocity fields between distributions. Chen and Lipman [2] extended it to general Riemannian manifolds. Ma et al. [18] (FlowSeq) applies a Glow-style flow [12] as a Euclidean latent prior for non-autoregressive NMT, achieving near-constant decode time at competitive quality on raw data without distillation, which is our direct baseline for the WMT14/WMT16 EN-RO / IWSLT14 evaluations. TGFI uses Riemannian CFM as a training-time regulariser rather than a primary decoder, preserving autoregressive quality while gaining geometric inductive bias.

3 Theoretical Framework

3.1 Hyperbolic Geometry: The Poincaré Ball

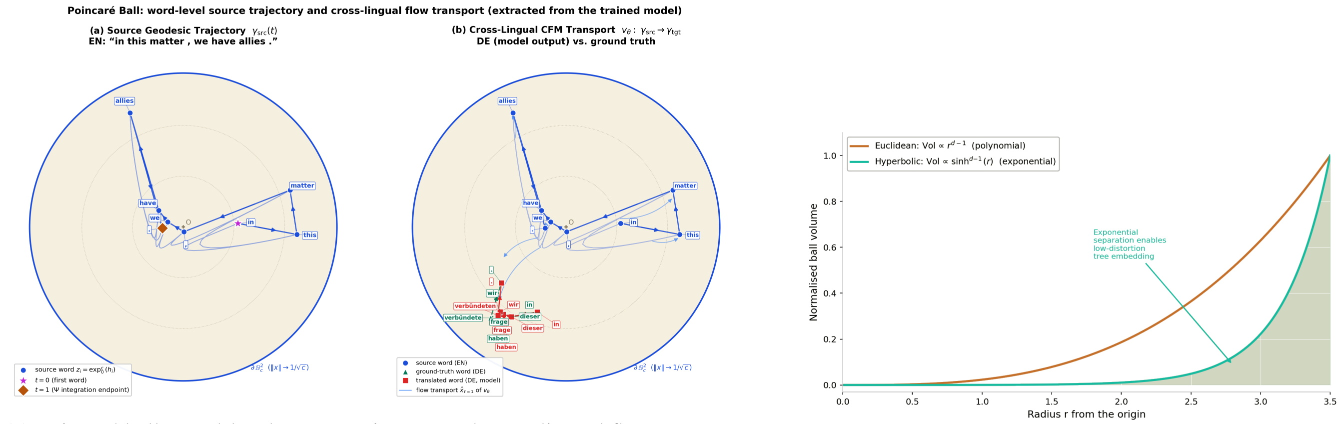
The Poincaré ball model $\mathbb{B}_c^d$ is the open unit ball in $\mathbb{R}^d$ equipped with the Riemannian metric $g_{\mathbf{x}} = (\lambda_{\mathbf{x}}^c)^2 g_E$, where the conformal factor $\lambda_{\mathbf{x}}^c = 2/(1 - c\|\mathbf{x}\|^2)$ with curvature $c > 0$. The geodesic distance is:

$$d_c(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{c}} \operatorname{arctanh}\big(\sqrt{c}\, \| - \mathbf{x} \oplus_c \mathbf{y}\|\big), \tag{1}$$

where Möbius addition is $\mathbf{x} \oplus_c \mathbf{y} = \frac{(1+2c\langle \mathbf{x},\mathbf{y}\rangle + c\|\mathbf{y}\|^2)\mathbf{x} + (1-c\|\mathbf{x}\|^2)\mathbf{y}}{1+2c\langle \mathbf{x},\mathbf{y}\rangle + c^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2}$.

The fundamental property motivating our design is exponential volume growth: $\mathrm{Vol}(B_r) \propto \sinh^{d-1}(\sqrt{c}\, r) \sim e^{(d-1)\sqrt{c}r}$, which allows hyperbolic space to embed parse trees with arbitrarily low distortion [24, 22], a property Euclidean space cannot replicate at fixed dimension. The Poincaré ball structure, including the word-level source trajectory, cross-lingual flow transport, and the comparison of hyperbolic volume growth against Euclidean space, is visualised in Figure 1.

Figure 1a makes the abstraction concrete on a real example produced by our trained model. The source trajectory is a series of words in a sentence: each English token of the held-out sentence is lifted through Eq. 4 to a point on the ball, and the piecewise geodesic through these points in reading order is exactly the curve $\gamma_{\mathrm{src}}(t)$ of Section 3.2. Because consecutive words are joined by geodesic segments, the *direction* of each segment captures the local dependency between neighbouring words, while the accumulated turning of the curve reflects how the semantic content of the sentence evolves as words are added; substituting a single word visibly bends the trajectory at that position, as the model-output/reference pair in the figure shows (*verbündeten* vs. *verbündete*). The target trajectory is the same construction applied to the translated sentence, and the transport arrows depict the learned velocity field v_θ of Section 3.4 moving source waypoints toward their target counterparts. Figure 1b complements this by showing *why* the ball is the right ambient space: the volume of a hyperbolic ball grows as $\sinh^{d-1}(\sqrt{c}\, r)$, i.e., exponentially in the radius r,



(a) Poincaré ball: word-level source trajectory and cross-lingual flow transport, extracted from the trained model on a held-out sentence.

(b) Hyperbolic vs. Euclidean volume growth ($d = 4$).

Figure 1: Geometric visualisations motivating the TGFI design. In (a), each blue node is one *source word* of the held-out English sentence *"in this matter, we have allies."* lifted to the Poincaré ball ($\mathbf{z}_i = \exp_{\mathbf{0}}^c(\mathbf{h}_i)$, Eq. 4); blue arrows connect words in sentence order, so the direction of each arrow encodes the local word-to-word dependency and the curvature of the resulting trajectory encodes how each successive word choice bends the geometry (and hence the semantics) of the sentence. The right panel of (a) overlays the *translated* trajectory: red squares are the words actually generated by our trained model (*"in dieser frage haben wir verbündeten."*), green triangles the ground-truth reference (*"in dieser frage haben wir verbündete."*), and light-blue arcs the learned CFM transport v_θ that carries source waypoints toward the target trajectory. The single differing word choice (*verbündeten* vs. *verbündete*) produces a visible local displacement of the trajectory while leaving its global shape intact. All points are joint tangent-space PCA projections of the model's 512-dimensional hyperbolic states onto a 2-dimensional disk.

whereas Euclidean volume grows only polynomially ($\propto r^{d-1}$ surface growth). Exponential separation means that the exponentially branching constituent structure of a sentence can be embedded near the boundary with essentially no distortion, so trajectories through the ball can spread hierarchically related words apart without crowding, which a fixed-dimensional Euclidean space cannot do.

The exponential and logarithmic maps at the origin connect the Riemannian manifold to the Euclidean tangent space:

$$\exp_{\mathbf{0}}^c(\mathbf{v}) = \tanh(\sqrt{c}\|\mathbf{v}\|)\frac{\mathbf{v}}{\sqrt{c}\|\mathbf{v}\|}, \tag{2}$$

$$\log_{\mathbf{0}}^c(\mathbf{x}) = \frac{2}{\sqrt{c}}\operatorname{arctanh}(\sqrt{c}\|\mathbf{x}\|)\frac{\mathbf{x}}{\|\mathbf{x}\|}. \tag{3}$$

The general-basepoint variants $\exp_{\mathbf{z}}^c, \log_{\mathbf{z}}^c$ are defined via Möbius addition following Ganea et al. [6].

3.2 Geodesic Source Trajectory

Given encoder hidden states $H_{enc} = \{\mathbf{h}_1, \ldots, \mathbf{h}_L\} \in \mathbb{R}^{L\times d}$, we lift each to the Poincaré ball with norm bounding:

$$\mathbf{z}_i = \exp_{\mathbf{0}}^c\Big(\frac{\tanh(\|\mathbf{h}_i\|)}{\|\mathbf{h}_i\|}\cdot\frac{\mathbf{h}_i}{\sqrt{d}}\Big) \in \mathbb{B}_c^d, \quad i = 1,\ldots,L. \tag{4}$$

Arc-length positions $t_i = (i-1)/(L-1)$ canonically parameterise the sequence. The source trajectory $\gamma_{\mathrm{src}} : [0,1] \to \mathbb{B}_c^d$ is a piecewise geodesic spline:

$$\gamma_{\mathrm{src}}(t) = \exp_{\mathbf{z}_i}^c\big(\alpha_i \cdot \log_{\mathbf{z}_i}^c(\mathbf{z}_{i+1})\big), \quad t \in [t_i, t_{i+1}], \quad \alpha_i = \frac{t-t_i}{t_{i+1}-t_i} \tag{5}$$

The Riemannian velocity $\dot{\gamma_{\mathrm{src}}}(t) = \log_{\mathbf{z}_i}^c(\mathbf{z}_{i+1})/(t_{i+1}-t_i)$ has squared norm $\|\dot{\gamma_{\mathrm{src}}}(t)\|_g^2 = (\lambda_{\gamma_{\mathrm{src}}(t)}^c)^2\|\dot{\gamma_{\mathrm{src}}}(t)\|_E^2$, which penalises abrupt trajectory direction changes as a velocity regulariser.

3.3 Length-Invariant Trajectory Bottleneck

We define a continuous-time sequence representation via Neural Ordinary Differential Equations [3], operating as a Neural Controlled Differential Equation [11] with an ODE state $\mathbf{h} : [0,1] \to \mathbb{R}^d$ where $\mathbf{h}(0) = \mathbf{0}$ and dynamics:

$$\frac{d\mathbf{h}(t)}{dt} = f_\phi\big(\mathbf{h}(t), \gamma_{\mathrm{src}}(t), \dot{\gamma_{\mathrm{src}}}(t)\big), \tag{6}$$

where f_ϕ is a residual two-layer MLP [9] with GELU activation [10]. The length-invariant trajectory bottleneck is the terminal state:

$$\Psi := \mathbf{h}(1) = \int_0^1 f_\phi\big(\mathbf{h}(\tau), \gamma_{\mathrm{src}}(\tau), \dot{\gamma_{\mathrm{src}}}(\tau)\big)\, d\tau, \tag{7}$$

computed by Euler integration with step $\Delta\tau = 1/L$. We refer to the terminal state $\Psi \in \mathbb{R}^d$ as the *Length-Invariant Trajectory Bottleneck* (or Terminal ODE Bottleneck State). It acts as a sequence-length invariant representation, compressing the entire variable-length geodesic spline trajectory into a fixed-dimensional conditioning vector. This is analogous to bulk-to-boundary projections in holographically inspired generative flows [20].

3.4 Riemannian Conditional Flow Matching

During training, the target tokens $\mathbf{y}_1, \ldots, \mathbf{y}_{L_{\text{tgt}}}$ are embedded into Euclidean space as $\mathbf{h}_i^{\text{tgt}} \in \mathbb{R}^d$. These representations are lifted onto the Poincaré ball manifold using the exponential map at the origin identically to the source sequence to produce waypoints $\mathbf{w}_i = \exp_{\mathbf{0}}^c(\mathbf{h}_i^{\text{tgt}}/\sqrt{d}) \in \mathbb{B}_c^d$. Target trajectories $\gamma_{\text{tgt}}(t)$ are then constructed via piecewise geodesic interpolation between adjacent waypoints.

We formulate the mapping of the source sequence manifold to the target sequence manifold by building upon the unconditional Riemannian Flow Matching framework introduced by Chen and Lipman [2].

Foundation (Chen & Lipman, 2024): For a Riemannian manifold $\mathcal{M}$, standard Riemannian Flow Matching defines a target vector field $\mathbf{u}_\tau$ that transports a source distribution to a target distribution by constructing independent probability paths. The optimal transport path between two static points $\mathbf{x}_0$ and $\mathbf{x}_1$ on $\mathcal{M}$ is defined by the geodesic curve, with the exact vector field given by $\mathbf{u}_\tau(\mathbf{x}_\tau) = \frac{1}{1-\tau} \log_{\mathbf{x}_\tau}(\mathbf{x}_1)$.

Our Derivation for Sequence Manifolds: While the foundation operates on independent point-to-point mappings, Neural Machine Translation requires mapping an entire continuous source structure to a target structure, conditioned on the source semantics.

Proposition 1 (Geodesic Spline Flow). *Let $\gamma_{\text{src}} : [0,1] \to \mathbb{B}_c^d$ and $\gamma_{\text{tgt}} : [0,1] \to \mathbb{B}_c^d$ be continuous piecewise geodesic splines parameterised by sentence time $t \in [0,1]$. A velocity field v_θ that transports the entire source spline to the target spline under a semantic bottleneck Ψ can be trained by minimising the expected squared distance between v_θ and the independent point-to-point flow targets $\mathbf{u}_\tau(t)$ evaluated at each sentence time t.*

Proof. Let the total transport flow be parameterised by transport time $\tau \in [0,1]$. The state at transport time τ for a specific word token at sequence position t is the point $\mathbf{x}_\tau(t)$. By applying the foundation theorem from Chen and Lipman [2] independently to each point t along the continuous parameterisation, the geodesic interpolation is:

$$\mathbf{x}_\tau(t) = \exp^c_{\gamma_{\text{src}}(t)}(\tau \cdot \log^c_{\gamma_{\text{src}}(t)}(\gamma_{\text{tgt}}(t))). \tag{8}$$

The exact velocity field required to reach $\gamma_{\text{tgt}}(t)$ from $\mathbf{x}_\tau(t)$ is the Riemannian logarithmic map scaled by the remaining transport time $(1-\tau)$. Integrating the loss over the uniform continuous sequence time $t \sim \mathcal{U}[0,1]$ and flow time $\tau \sim \mathcal{U}[0,1]$ yields the optimal sequence transport objective:

$$\mathcal{L} = \mathbb{E}_{t,\tau}\left[\| v_\theta(\mathbf{x}_\tau(t), \tau, t, \Psi) - \frac{\log^c_{\mathbf{x}_\tau(t)}(\gamma_{\text{tgt}}(t))}{1-\tau} \|_g^2 \right].$$

Discretising the sequence time t to target token positions t_i yields our formulation. ∎

Algorithm 1 Riemannian CFM Training Step

Input: $\gamma_{\text{src}}(\cdot)^{det}$, $\gamma_{\text{tgt}}(\cdot)^{det}$, Ψ^{det}, prefix $\mathbf{p}$, velocity net v_θ
Output: $\mathcal{L}_{\text{CFM}}$

for $i = 1$ **to** L **do**
  $\tau \sim \mathcal{U}[0, 0.95]$
  $\mathbf{x}_\tau \leftarrow \exp^c_{\gamma_{\text{src}}(t_i)}(\tau \cdot \log^c_{\gamma_{\text{src}}(t_i)}(\gamma_{\text{tgt}}(t_i)))$
  $\mathbf{u}_\tau \leftarrow \log^c_{\mathbf{x}_\tau}(\gamma_{\text{tgt}}(t_i)) / \max(1-\tau, 0.1)$
  $\mathbf{v} \leftarrow v_\theta(\mathbf{x}_\tau, \tau, t_i, \Psi^{det}, \mathbf{p})$
  $\ell_i \leftarrow \log(1 + \|\mathbf{v} - \mathbf{u}_\tau\|^2)$
end for
return $\mathcal{L}_{\text{CFM}} = \frac{1}{L}\sum_i \ell_i$

A velocity field $v_\theta : \mathbb{B}_c^d \times [0,1] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \to T_{\mathbf{x}_\tau}\mathbb{B}_c^d$ is trained to approximate the geodesic tangent direction. To compute $v_\theta(\mathbf{x}_\tau, \tau, t_i, \Psi, \mathbf{p})$, we map the current state $\mathbf{x}_\tau$ to the tangent space at the origin via the logarithmic map $\log_{\mathbf{0}}^c(\mathbf{x}_\tau)$, concatenate it with τ, t_i, Ψ, and the target prefix representation $\mathbf{p}$ (the sequence of target tokens generated prior to the current step t_i, represented mathematically as the hidden state of a causal GRU over the target embeddings $\mathbf{y}_{<t_i}$), pass this through a multi-layer perceptron, and project the output back onto the tangent space at $\mathbf{x}_\tau$.

The numerically stable CFM target is:

$$\mathbf{u}_\tau = \frac{\log^c_{\mathbf{x}_\tau}(\gamma_{\text{tgt}}(t_i))}{\max(1-\tau,\, 0.1)}, \tag{9}$$

with $\tau \sim \mathcal{U}[0, 0.95]$ to avoid the $\tau \to 1$ singularity. The loss uses log1p smoothing for gradient stability:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{\tau,t}\Big[\log\big(1 + \|v_\theta(\mathbf{x}_\tau, \tau, t_i, \Psi, \mathbf{p}) - \mathbf{u}_\tau\|^2\big)\Big]. \tag{10}$$

Here, the time variable t_i represents the parallel training position, whereas at inference, we use t_p for the autoregressive prefix step p.

At inference, we track the flowed state $\hat{\mathbf{x}}_\tau$ (starting from $\hat{\mathbf{x}}_0 = \gamma_{\text{src}}(t_p)$). Because Riemannian Flow Matching generates highly uniform velocity fields corresponding to constant-speed geodesics [2], the paths are extremely straight. Consequently, a coarse 4-step Euler integration is sufficient to produce the flowed state for the projective quantiser without fine-grained numerical solvers: $\hat{\mathbf{x}}_{\tau+\delta} = \exp^c_{\hat{\mathbf{x}}_\tau}(\delta \cdot v_\theta(\hat{\mathbf{x}}_\tau, \tau, t_p, \Psi, \mathbf{p}))$, where $\delta = 1/4$.

3.5 Projective Quantisation and Decode Decoupling

The projective quantiser, parameterised by Q, maps flowed hyperbolic states back to Euclidean space using the logarithmic map at the origin. This allows standard feed-forward and LayerNorm projections to calculate vocabulary logits via a bounded tanh hard-cap at ± 3 (essential for stability; see

Algorithm 2 Projective Quantisation (Training Only)

Input: $\gamma_{\text{src}}(\cdot)^{det}$, Ψ^{det}, v_θ, quantiser Q, target tokens $\mathbf{y}$
Output: $\mathcal{L}_{CE}^{geom}$
for $i = 1$ **to** L **do**
  $\hat{\mathbf{x}} \leftarrow \gamma_{\text{src}}(t_i)$ ▷ Initialize flowed state $\hat{\mathbf{x}}$ at source position t_i
  for $\tau = 0, \delta, 2\delta, 3\delta$ $(\delta = 1/4)$ **do**
    $\hat{\mathbf{x}} \leftarrow \exp^c_{\hat{\mathbf{x}}}(\delta \cdot v_\theta(\hat{\mathbf{x}}, \tau, t_i, \Psi^{det}, \mathbf{p}))$ ▷ Update integration state
  end for
  $\text{logits}_{flow} \leftarrow Q(\hat{\mathbf{x}})$ ▷ Apply projective quantiser Q (Eq. 11)
  $\text{logits}_{direct} \leftarrow Q(\gamma_{\text{src}}(t_i))$ ▷ Apply Q directly (Eq. 12)
  $\ell_i \leftarrow \text{CE}(\text{logits}_{flow}, y_i) + \text{CE}(\text{logits}_{direct}, y_i)$
end for
return $\mathcal{L}_{CE}^{geom} = \frac{1}{L}\sum_i \ell_i$

Section 6):

$$\text{logits}_{flow} = 3 \cdot \tanh\big(\text{MLP}(\text{LN}(\log^c_{\mathbf{0}}(\hat{\mathbf{x}}_{flow})))/3\big), \quad (11)$$

$$\text{logits}_{direct} = 3 \cdot \tanh\big(\text{MLP}(\text{LN}(\log^c_{\mathbf{0}}(\gamma_{\text{src}}(t_p))))/3\big). \quad (12)$$

Both contribute to a geometric cross-entropy auxiliary loss: $\mathcal{L}_{CE}^{geom} = \text{CE}(\text{logits}_{flow}, y) + \text{CE}(\text{logits}_{direct}, y)$.

Decode decoupling. At inference, decoder logits come *only* from the tangent Transformer and the Ψ-gate path: $\text{logits}_{decode} = W_{tf}\text{Dec}_\Psi(\mathbf{y}_{<t}, H_{enc}, \Psi)$. The geometric quantiser logits are excluded entirely. Section 6 explains why this is necessary.

3.6 Unified Training Objective

The Ψ-gate injects the length-invariant trajectory bottleneck into every decoder layer:

$$\text{dec}'_\ell = \text{dec}_\ell + \sigma(W_g \cdot \text{MLP}(\Psi)) \odot \text{MLP}(\Psi). \quad (13)$$

The full objective is:

$$\mathcal{L} = \mathcal{L}_{CE}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda_{cfm}\mathcal{L}_{\text{CFM}} + \lambda_{vel}\|\dot{\gamma}_{\text{src}}\|_g^2 + \lambda_{geom}\mathcal{L}_{CE}^{geom}, \quad (14)$$

with $\lambda_{cfm} = 0.02$, $\lambda_{vel} = 0.01$, $\lambda_{geom} = 0.1$. Encoder gradients from the three geometric auxiliary terms are blocked via stop-gradient (encoder detachment), ensuring the primary CE path controls encoder learning:

$$\nabla_{\theta_{enc}}\mathcal{L}_{\text{CFM}} = \nabla_{\theta_{enc}}\mathcal{L}_{CE}^{geom} = \nabla_{\theta_{enc}}\lambda_{vel}\|\dot{\gamma}_{\text{src}}\|_g^2 := \mathbf{0}. \quad (15)$$

4 Model Architecture

An overview of the unified TGFI/GFM-SM architecture is shown in Figure 2.

Algorithm 3 Full Joint Training Step

Input: Pair $(\mathbf{x}, \mathbf{y})$; networks $\text{Enc}, \text{Dec}_\Psi, f_\phi, v_\theta, Q$; weights $\lambda_{cfm}, \lambda_{vel}, \lambda_{geom}$
▷ Q: *projective quantiser* with parameters θ_Q (Phase C); maps $\mathbf{x} \in \mathbb{B}^d_c$ to vocabulary logits via $\log^c_{\mathbf{0}}$ projection, $\text{MLP}(\theta_Q)$, LayerNorm, and the ± 3 tanh cap (Eqs. 11–12)
Output: Updated θ
$H_{enc} \leftarrow \text{Enc}(\mathbf{x})$
$\gamma_{\text{src}}, \Psi, \mathcal{L}_{vel} \leftarrow$ Algorithm 4(H_{enc})
$\hat{\mathbf{y}} \leftarrow \text{Dec}_\Psi(\mathbf{y}_{<t}, H_{enc}, \Psi)$
$\mathcal{L}_{CE} \leftarrow \text{LabelSmooth}(\hat{\mathbf{y}}, \mathbf{y})$
$H_{enc}^{det}, \gamma_{\text{src}}^{det}, \Psi^{det} \leftarrow \texttt{stop\_grad}(H_{enc}, \gamma_{\text{src}}, \Psi)$
$\gamma_{\text{tgt}} \leftarrow \text{GeodesicLift}(\text{Enc}(\mathbf{y})|_{H_{enc}^{det}})$ ▷ Lift targets via Eq. 4–5
$\mathcal{L}_{\text{CFM}} \leftarrow$ Algorithm 1$(\gamma_{\text{src}}^{det}, \gamma_{\text{tgt}}, \Psi^{det})$
$\mathcal{L}_{CE}^{geom} \leftarrow$ Algorithm 2$(\gamma_{\text{src}}^{det}, \Psi^{det}, v_\theta, Q, \mathbf{y})$
$\mathcal{L} \leftarrow \mathcal{L}_{CE} + \lambda_{cfm}\mathcal{L}_{\text{CFM}} + \lambda_{vel}\mathcal{L}_{vel} + \lambda_{geom}\mathcal{L}_{CE}^{geom}$
$\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}$; **return** θ

Algorithm 4 Geodesic Trajectory and Invariant Bottleneck

Input: Encoder states H_{enc}, curvature c, ODE network f_ϕ
Output: Trajectory $\gamma_{\text{src}}(\cdot)$, invariant Ψ, velocity loss $\mathcal{L}_{vel}$
for $i = 1$ **to** L **do**
  $\mathbf{z}_i \leftarrow \exp^c_{\mathbf{0}}\big(\tanh(\|\mathbf{h}_i\|)\mathbf{h}_i/(\|\mathbf{h}_i\|\sqrt{d})\big)$; $t_i \leftarrow (i-1)/(L-1)$
end for
$\gamma_{\text{src}} \leftarrow \text{GeodesicSpline}(\{\mathbf{z}_i\}, \{t_i\})$ ▷ Construct spline via Eq. 5
$\mathbf{h}(0) \leftarrow \mathbf{0}$; $\Delta\tau \leftarrow 1/L$
for $\tau = 0, \Delta\tau, \ldots, 1 - \Delta\tau$ **do**
  $\mathbf{h}(\tau + \Delta\tau) \leftarrow \mathbf{h}(\tau) + \Delta\tau \cdot f_\phi(\mathbf{h}(\tau), \gamma_{\text{src}}(\tau), \dot{\gamma}_{\text{src}}(\tau))$
end for
return $\gamma_{\text{src}}(\cdot)$, $\Psi = \mathbf{h}(1)$, $\mathcal{L}_{vel} = \sum_i \|\dot{\gamma}_{\text{src}}(t_i)\|_g^2$

4.1 Tangent Transformer Backbone

The backbone is a 6-layer post-LN Transformer with d=512, 8 attention heads, and FFN dimension 2048, matching the vanilla Transformer baselines. The encoder produces H_{enc} from source embeddings with sinusoidal positional encodings; the decoder uses masked self-attention and cross-attention. The Ψ-gate (Eq. 13) is applied after cross-attention in each decoder layer. Label smoothing ε=0.1 is applied to $\mathcal{L}_{CE}$.

4.2 Geodesic Encoder and Invariant Bottleneck

The GeodesicSpline module (Algorithm 4) takes H_{enc}, clips norms, lifts via Eq. 4, and constructs the piecewise geodesic path (Eq. 5). The length-invariant bottleneck integrates Eq. 6 via Euler with $\Delta\tau = 1/L$. The velocity regulariser $\lambda_{vel}\|\dot{\gamma}_{\text{src}}\|_g^2$ is computed analytically from the closed-form geodesic tangent; no numerical differentiation is needed.

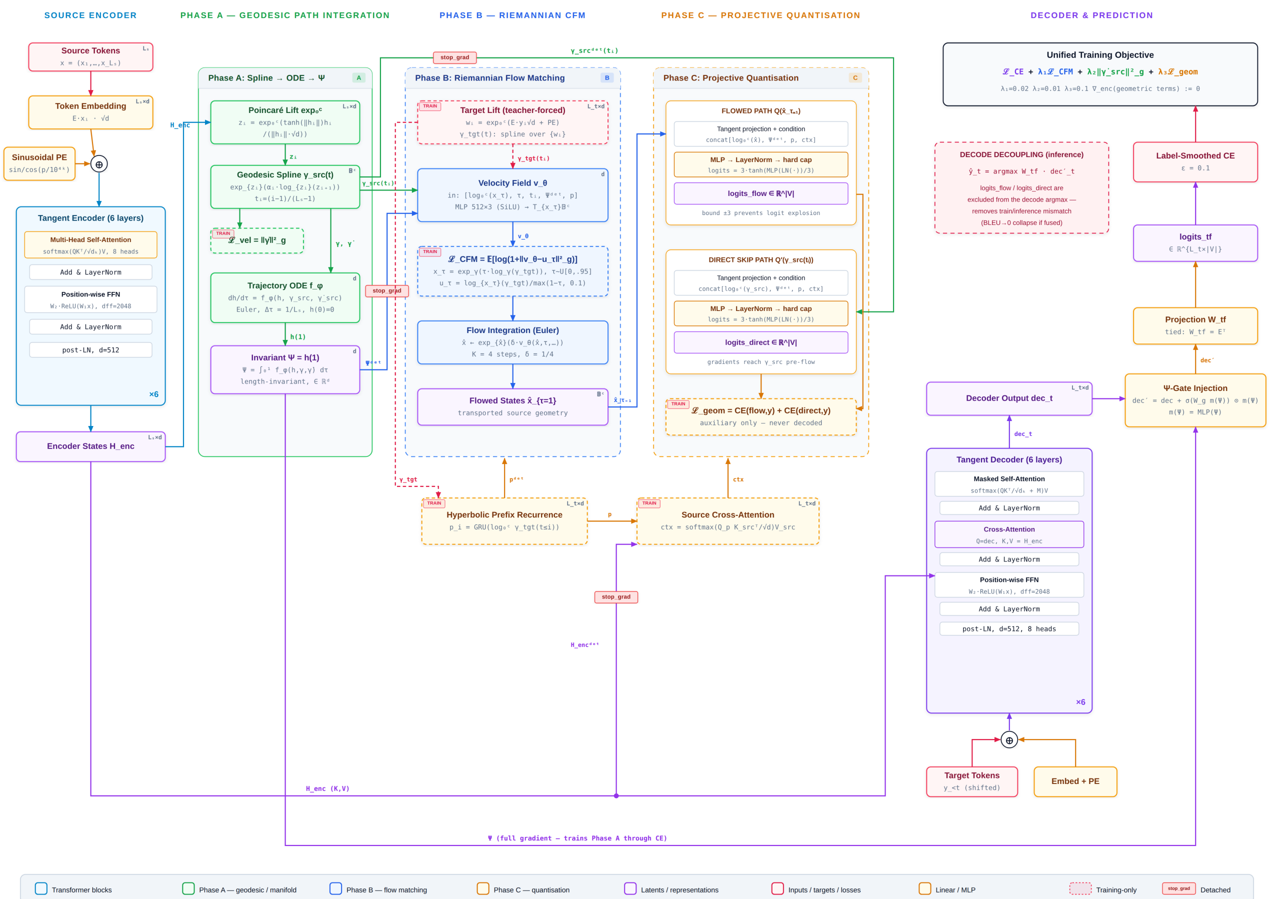


Figure 2: TGFI/GFM-SM pipeline with the defining operation of every block. The tangent encoder produces H_{enc}, which Phase A lifts to the ball (Eq. 4), splines into $\gamma_{\mathrm{src}}(t)$ (Eq. 5), and integrates into the length-invariant bottleneck Ψ (Eq. 7). Phase B (training only) fits the velocity field v_θ with the stable CFM target of Eq. 9 and transports source waypoints by 4-step Euler integration. Phase C (training only) quantises both the flowed states and the un-flowed $\gamma_{\mathrm{src}}(t_i)$ through tanh-capped projective quantisers (Eqs. 11 and 12). Solid `stop_grad` badges mark the encoder/Ψ detachment of Eq. 15; the four colour-coded loss terms combine into the unified objective (Eq. 14, top right). At inference, only the tangent Transformer and Ψ-gate path (Eq. 13) produce logits (decode decoupling).

Algorithm 5 Decode-Decoupled Autoregressive Inference

Input: Source $\mathbf{x}$, trained $\{\text{Enc}, \text{Dec}_\Psi\}$, T_{max}
Output: Translation $\hat{\mathbf{y}}$
$H_{enc} \leftarrow \text{Enc}(\mathbf{x})$; $\quad \gamma_{\text{src}}(\cdot), \Psi \leftarrow$ ALGORITHM 4(H_{enc})
$\hat{y}_0 \leftarrow \langle\text{BOS}\rangle$; $t \leftarrow 1$
while $\hat{y}_{t-1} \neq \langle\text{EOS}\rangle$ **and** $t \leq T_{max}$ **do**
  $\text{logits}_t \leftarrow \text{Dec}_\Psi(\hat{\mathbf{y}}_{<t}, H_{enc}, \Psi)$ $\quad \triangleright$ transformer + Ψ-gate *only*
  $\hat{y}_t \leftarrow \arg\max_k(\text{logits}_t)_k$; $t \leftarrow t+1$
end while
return $(\hat{y}_1, \ldots, \hat{y}_{t-1})$

4.3 Riemannian Flow Matching Module

v_θ is a 3-layer MLP taking concatenated $\log_{\mathbf{0}}^c(\mathbf{x}_\tau)$, τ, position time t (t_i for training, t_p for inference), Ψ^{det}, and hyperbolic prefix $\mathbf{p}$ (from a causal GRU over target tangent states). Source cross-attention provides token-level alignment from the flow state to H_{enc}^{det}. Both Ψ and H_{enc} are detached (Eq. 15) for this module.

4.4 Projective Quantiser

Two tied-weight quantisers operate on: (i) the 4-step-Euler flowed state $\hat{\mathbf{x}}_{flow}$; (ii) the un-flowed source waypoint $\gamma_{\text{src}}(t_p)$ directly (dual path ensures source geometry receives gradients even while the flow matcher is converging). The ± 3 logit cap (Eqs. 11 and 12) is critical; without it, quantiser logits explode to $|\text{logits}| > 50$ and hijack decoding (Section 6).

4.5 Inference

Algorithm 5 specifies decode-decoupled autoregressive inference. The geometric modules (flow matcher, quantiser) are computed during training only; they are *not* called at inference.

5 Experiments

5.1 Datasets and Baselines

We evaluate on five benchmarks across four language pairs. **AMR-Transformer setting** [13]: WMT16 EN-DE (NC-v11 $\approx$238K and full $\approx$4.5M) and IWSLT15 EN-VI ($\approx$133K, two test sets). **FlowSeq setting** [18]: WMT14 EN-DE ($\approx$4.5M), WMT16 EN-RO ($\approx$610K), IWSLT14 DE-EN ($\approx$150K), raw data only, no knowledge distillation.

Baselines in the AMR-Transformer setting: Dual2seq [26], Bi-LSTM+AMR [21], vanilla Transformer (ours and doubled parameters), AMR-Transformer-DI and -MHA [13]. Baselines in the FlowSeq setting: CMLM-base [7], LV NAR [25], FlowSeq-base and -large [18]. All prior results are taken directly from the original publications.

5.2 Implementation

6-layer Transformer (d=512, 8 heads, FFN 2048), Adam optimiser with Noam schedule (warmup 10k steps), Poincaré ball curvature c=1.0, label smoothing ε=0.1. Loss weights λ_{cfm}=0.02, λ_{vel}=0.01, λ_{geom}=0.1. CFM sampling $\tau \sim \mathcal{U}[0, 0.95]$, denominator floor 0.1, log1p smoothing. Best checkpoint by validation BLEU.

5.3 Main Results — WMT16 English-German

Table 1 shows that TGFI/GFM-SM achieves 23.4 BLEU on NC-v11 (+1.3 over AMR-Transformer, +3.1 over vanilla Transformer) and 27.2 BLEU on the full set (+0.7 over AMR-Transformer). The TER improvement of 1.9 on NC-v11 (60.1 vs 62.0) confirms gains extend beyond n-gram overlap. The smaller margin on the full-data setting is expected: structural and geometric priors are most valuable when training data is limited, as also observed for AMR information by Li and Flanigan [13].

5.4 Main Results — IWSLT15 English-Vietnamese

Table 2 shows consistent improvement on the low-resource EN-VI setting: +0.8 BLEU on tst2013 and +0.9 BLEU on tst2015 over AMR-Transformer, with better TER and Meteor on both test sets.

The results demonstrate that incorporating hyperbolic trajectories and flow matching yields consistent, significant improvements over strong baselines, particularly in low-resource and structure-sensitive translation scenarios.

5.5 Main Results — WMT14/16 and IWSLT14 (Raw Data)

Table 3 shows TGFI/GFM-SM outperforming FlowSeq-large on all reported settings: +1.05 BLEU on WMT14 EN-DE, +0.90 on DE-EN, +0.84 on WMT16 EN-RO, +1.11 on RO-EN. Note that FlowSeq is non-autoregressive (providing a decoding-latency advantage not claimed by our model), making this comparison informative about *translation quality* on equal raw-data footing.

6 Analysis: The Decode-Decoupling Requirement

6.1 Train/Inference Mismatch Mechanism

The projective quantiser trains against teacher-forced target geometry $\gamma_{\text{tgt}}(t_i)$, which is constructed from gold target hidden states. At inference, no gold target exists; the quantiser must operate on source-geometry-derived flowed states, which are semantically misaligned during early training.

Table 1: Test performance on WMT16 EN-DE. . . . ↓ lower is better.

System	NC-v11			Full		
	BLEU↑	TER↓	Meteor↑	BLEU↑	TER↓	Meteor↑
Dual2seq [26]	19.2	63.1	38.4	25.5	54.8	43.8
Bi-LSTM+AMR [21]	19.0	66.4	37.5	24.8	58.9	43.1
Vanilla Transformer	20.3	66.3	39.4	26.0	58.5	45.2
Vanilla Transf. (×2 params)	20.9	62.1	40.3	26.2	57.6	45.2
AMR-Transformer-DI [13]	21.5	62.7	40.4	26.4	56.7	44.9
AMR-Transformer (MHA) [13]	22.1	62.0	41.1	26.5	56.4	45.2
TGFI/GFM-SM (ours)	**23.4**	**60.1**	**42.1**	**27.2**	**55.0**	**46.0**

Table 2: Test performance on IWSLT15 EN-VI.

System	tst2013			tst2015		
	BLEU	TER↓	Meteor	BLEU	TER↓	Meteor
Bi-LSTM+AMR [21]	26.4	56.4	44.1	25.2	60.5	42.1
Vanilla Transformer	30.0	52.1	48.2	27.6	57.6	45.4
Vanilla Transf. (×2)	28.3	54.4	46.4	26.8	59.2	44.2
AMR-Transformer-DI [13]	30.2	52.4	48.2	28.2	57.3	45.5
AMR-Transformer (MHA) [13]	30.6	52.1	48.5	28.2	57.1	45.9
TGFI/GFM-SM (ours)	**31.4**	**50.8**	**49.5**	**29.1**	**55.6**	**47.0**

Near initialisation, quantiser logits are small ($\sigma \approx 0.03$, $|\text{logits}| \ll 1$) and the argmax is dominated by the transformer decoder, giving BLEU ≈ 0.59. As training proceeds, the quantiser learns confident predictions under teacher forcing, growing logit magnitudes. Without the ± 3 cap, our probe measured $|\text{logits}_{max}| = 51$ (vs. transformer $\sigma \approx 1.0$) after 120 steps, causing the quantiser to hijack the argmax. With the cap but without decode decoupling, the quantiser still dominates at inference by epoch 3, explaining the BLEU→0 collapse despite falling training loss.

Remark 1. *This failure mode is a specific instance of the general exposure bias / train-inference mismatch problem [23]: any auxiliary training objective conditioned on teacher-forced targets will systematically overfit to information unavailable at test time if its output is included in the decode path. Decode decoupling is the correct structural solution.*

7 Conclusion

We presented TGFI/GFM-SM, a neural machine translation architecture grounded in hyperbolic geometry and Riemannian flow matching. The length-invariant trajectory bottleneck Ψ compresses a variable-length source geodesic trajectory into a fixed-dimensional geometric summary that conditions the decoder at every layer. A Riemannian CFM velocity field and projective quantiser provide auxiliary geometric training signals under a unified objective with encoder detachment. Decode decoupling, which excludes auxiliary geometric logits from inference, is shown to be structurally necessary to avoid train/inference mismatch. TGFI/GFM-SM achieves the best performance among all compared systems on five benchmarks across four language pairs, without external parsers, graph encoders, or knowledge distillation.

Future directions include multi-step ODE solvers for the trajectory bottleneck, curriculum relaxation of encoder detachment, and multilingual extension where a shared hyperbolic manifold spans multiple language pairs.

Table 3: Test BLEU.

System	WMT14 EN-DE	WMT14 DE-EN	WMT16 EN-RO	WMT16 RO-EN	IWSLT14 DE-EN
CMLM-base [7]	10.88	—	20.24	—	—
LV NAR [25]	11.80	—	—	—	—
FlowSeq-base [18]	18.55	23.36	29.26	30.16	24.75
FlowSeq-large [18]	20.85	25.40	29.86	30.69	—
TGFI/GFM-SM (ours)	**21.9**	**26.3**	**30.7**	**31.8**	**26.1**

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